

SHAKYA BHATTACHARYYA

New York, NY (ready to relocate)

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[Website](#)

SUMMARY

Business-focused Data Scientist with experience in predictive modeling, NLP, and decision-support dashboards. Skilled at structuring messy business problems, transforming datasets, and building tools that help teams make faster, clearer decisions. Brings experience in marketing analytics, retention modeling, search relevance, and logistics, with strong ability in Python, SQL, and BI tools.

PROFESSIONAL EXPERIENCE

Simon Business School

Jan 2025 - Present

Instructional Analytics Associate - Analytics Design & Application

Rochester, NY

- Led weekly hands-on sessions for 100+ MS students, helping them diagnose data issues, optimize SQL/Python workflows, and structure analytics projects the way real teams do.
- Designed a logistics optimization case study (for an anonymous client), guiding students through ETL, feature engineering, and BI dashboards that mirrored real operational decision-making.
- Built automated Python-based grading and feedback tools, reducing evaluation time by 70% while improving consistency and clarity.

Simon Vision Consulting

Sep 2024 - Dec 2024

Data Science Analyst (Client Engagement: City of Rochester)

Rochester, NY

- Developed retention models (Python, scikit-learn) improving AUC by 0.07 and surfacing key drivers for a projected 15% lift.
- Conducted controlled A/B tests and uplift modeling to evaluate program changes, summarizing findings for leadership and marketing teams to guide strategy.
- Used NLP (TF-IDF + topic modeling) on survey responses to surface dropout themes that shaped policy and funding proposals.

Cognizant Technology Solutions

Sep 2022 - Jun 2024

Business Data Analyst (ML Focus)

Kolkata, India

- Built ML forecasting and cost-risk models (XGBoost, regression) on 250K+ clinical trial records; cut decision turnaround by 20% and supported \$30M+ cost avoidance.
- Automated ETL and batch scoring workflows (SQL, cron jobs), reducing reporting cycles by 90% and improving delivery reliability.
- Deployed TensorFlow-based trial-risk models that increased detection precision by 12% and improved operational triage.
- Enhanced feature engineering and ML pipelines, improving efficiency by 20% and reducing latency across production runs.
- Partnered with cross-functional teams to translate modeling output into clearer resourcing and fraud-risk decisions.

IEM-IEDC (Govt. of India funded)

Feb 2019 - Dec 2021

Data Science Intern

Kolkata, India

- Improved model accuracy by 25% on 100K+ records using TensorFlow and scikit-learn in applied AI research settings.
- Applied NLP methods on case studies and delivered 10+ analytics workshops to 1,000+ learners on SQL, Python, and Tableau.

PROJECTS

Semantic Search Engine for Housing Listings | [Project Link](#)

Jan 2025 - Apr 2025

- Built a semantic search engine (TF-IDF and BERT embeddings) for 30K+ listings, improving retrieval relevance by 34%.
- Delivered a working prototype with keyword + semantic matching and similarity-ranked recommendations.

EDUCATION

Simon Business School, University of Rochester

Jul 2024 - Dec 2025

Master of Science, Business Analytics (STEM)

Rochester, NY

- **GPA:** 3.9/4.0
- **Achievements:** Advanced Certificate in Pricing Analytics (Revenue Optimization Focus)

Maulana Abul Kalam Azad University of Technology

Aug 2018 - Jun 2022

Bachelor of Technology, Computer Science & Engineering (STEM)

India

- **GPA:** 8.24/10 (Top 10%)
- **Achievements:** Director's Award for Overall Student Performance

SKILLS

- **Programming & ML:** Python (pandas, numpy, scikit-learn, TensorFlow, XGBoost), SQL, Git
- **Analytics:** Predictive Modeling, NLP (TF-IDF, BERT), Forecasting, A/B Testing, Credit Risk Modeling, Time-series, Uplift Modeling, Clustering, Classification, Regression
- **Data Engineering:** ETL, Data Cleaning, Feature Engineering, Batch Scoring, API Deployment, AWS S3, Azure
- **Business Intelligence:** Tableau, PowerBI, Dashboard Design, KPI Reporting, Stakeholder Communication
- **Tools & Platforms:** Jupyter, VS Code, Excel, Docker, Postman, PostgreSQL, MySQL